Indirect reciprocity with inertial reputation updating

Introduction, Storyline 2: agreement from individual integration of evidence

Cooperation among unrelated individuals is sustained in part by reputation[1,2]. By helping others, individuals improve their standing and become more likely to receive help from third parties, a mechanism known as indirect reciprocity[3–6]. Theory has identified social norms that make reputation-based cooperation evolutionarily stable[7,8], most notably the leading-eight norms, which sustain cooperation when reputations are public, even in the presence of occasional errors[8,9]. These results rest on agreement about who is good and who is bad. When reputations are private and assessments are noisy, individuals hold different views of the same person[10,11], and most leading-eight norms no longer maintain stable cooperation[12–15]. A central problem for indirect reciprocity is therefore how reputations can remain both accurate and shared under private assessment. They must distinguish those who cooperate from those who do not, and different observers must largely agree about who is who[16,17].

The difficulty is sharpest for the norms that sustain cooperation under public information. Under these norms, a discriminator who refuses to help an individual in bad standing keeps its own good standing, because the refusal counts as justified punishment[7,8,18]. This exemption works only if the observer shares the donor's view of the recipient, which under private assessment is often not the case. An observer who still regards the recipient as good sees an unjustified refusal and condemns the discriminator. Disagreement about one individual thereby becomes disagreement about those who act towards it, and it spreads through the population[12,19,20]. Discriminators thus face a punishment dilemma. Withholding help from defectors protects cooperation against exploitation, yet each act of punishment risks their own reputation in the eyes of observers who disagree[18,21,22]. A remedy must therefore keep justified punishers in good standing without letting genuine defectors escape condemnation.

Most proposed remedies keep punishers in good standing by aligning what observers know, because cooperation is stable only when individual opinions are sufficiently correlated[17]. Public institutions broadcast a single verdict[23], gossip spreads opinions between peers[24,25], and other rules synchronize private opinions directly[17] or evaluate actions from the donor's perspective[20]. Each route rests on conditions that are often unmet. Institutions need monitoring capacity and the adherence of those they serve[23], gossip must be frequent and accurate enough to produce consensus[25,26], and empathetic evaluation requires knowing how the donor sees the recipient[20]. These conditions are strained where empathy, communication and institutional trust are compromised, as in polarized societies[21,27]. Recent models therefore seek agreement from private information alone, by letting each observer judge from several fresh observations of the same individual[21] or assess recipients as well as donors[28]. Both approaches enlarge the evidence behind each judgement. Whether agreement can also arise from how individuals integrate the evidence they already receive, one observation at a time, remains open. In principle it can, because independent observers agree whenever each reaches a reliable verdict on its own. A rule that makes verdicts reliable by discounting contrary evidence, however, may also discount genuine defection.

Here, we develop a model of indirect reciprocity in which each observer integrates its own observations over time, without sharing information and without observing more than in standard private-assessment models (Fig. 1). Every observer keeps a private record of each individual. Negative assessments move an individual in good standing through a sequence of warning states before the observer's verdict turns bad, and positive assessments move an individual in bad standing through recovery states before it turns good. The two paths can differ in length, so that an observer may require more evidence to condemn than

to rehabilitate, or the reverse. Each intermediate state is interpreted as good or bad with a probability that sets how strongly incomplete evidence already affects the observer's behaviour.

We find that private observers can agree with one another and still distinguish cooperators from defectors, although they share no information, provided that inertia is allocated in a norm-specific direction. Agreement alone is not sufficient. With the same small inertia in both directions, observers under Shunning agree only because they regard nearly everyone as bad. Compact architectures, selected solely by these two reputational criteria, make cooperation stable and easy to reach under each of four classical norms (Scoring, Shunning, Stern Judging and Simple Standing). At the same depth, inertia placed on the wrong path can leave no cooperation at all, so how evidence is processed matters more than how much of it is retained. Finally, the tolerance that makes private verdicts robust can also shelter opportunists who defect only occasionally, a vulnerability that the aggregation and score-based rules we test share. Allowing warning states to carry partial consequences closes this loophole for the opportunistic strategies we consider, while cooperation remains high. Consensus about reputations therefore need not be imposed through shared information. It can emerge from how each individual weighs its own evidence.

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